

Flavonoids, a ubiquitous dietary phenolic subclass, exert extensive in vitro anti-invasive and in vivo anti-metastatic activities.

Cancer metastasis refers to the spread of cancer cells from the primary neoplasm to distant sites, where secondary tumors are formed, and is the major cause of death from cancer. Natural phytochemicals containing phenolic compounds have been widely demonstrated to have the capability to prevent cancer metastasis. Among phenolic compounds, flavonoids are a very large subclass, and they are abundant in food and nutraceuticals. The number of reports demonstrating that flavonoids are an effective natural inhibitor of cancer invasion and metastasis is increasing in the scientific literature. Catechin derivatives, (–)-epigallocatechin-3-gallate, (–)-epigallocatechin, (–)-epicatechin-3-gallate, and (–)-epicatechin, are the most studied compounds in this topic so far; genistein/genistin, silibinin, quercetin, and anthocyanin have also been widely investigated for their inhibitory activities on invasion/metastasis. Other flavonoids in dietary vegetable foods that are responsible for anti-invasive and anti-metastatic activities of tumors include luteolin, apigenin, myricetin, tangeretin, kaempferol, glycitein, licoricidin, daidzein, and naringenin. To effectively overcome the metastatic cascade, including cell-cell attachment, tissue barrier degradation, migration, invasion, cell-matrix adhesion, and angiogenesis, it is essential that a bioactive compound prevent tumor cells from metastasizing. This review summarizes the effects of flavonoids on the metastatic cascade and the related proteins, the in vitro anti-invasive activity of flavonoids against cancer cells, and the effects of flavonoids on antiangiogenic and in vivo anti-metastatic models. The available scientific evidence indicates that flavonoids are a ubiquitous dietary phenolics subclass and exert extensive in vitro anti-invasive and in vivo anti-metastatic activities.